PROJECT HANDOFF

Claude of Duty

A grounded first-person tactical shooter: physical weapon weight, real bore-axis ballistics, picture-in-picture optics, sound that travels at the speed of sound, and enemies that fight as fire teams.

Date	16 October 2026
Engine	Unity 6000.6.4f1 · HDRP 17.6.0 · Input System 1.20.0 · AI Navigation 2.0.14
Repository	github.com/LUK3N4Z/Claude-of-duty · branch `main` · head `da49e6e`
Status	All 16 milestones plus playtest fixes · 165 PlayMode tests (160 pass, 5 opt-in skipped, 0 fail)
Build	Windows 64-bit player (builder 9) boots into the Armory hub; the deployment table reaches every mode

Everything in the game — models, materials, weapons, parts, grenades, soldiers, sounds, voices, scenes and NavMeshes — is generated by editor tools. Models are high-detail procedural geometry with weathered PBR finishes until meshy.ai exports are dropped into `Art/Models`; gunfire is physically synthesized at 48 kHz and any layer can be replaced by a recording.

Contents

1 Status at a glance
2 Getting started
3 Controls
4 Movement, traversal and the player's body
5 Weapon handling, aiming and recoil
6 Weapon mechanics and weapons
7 Gunsmith and attachments
8 Ballistics, damage and hit confirmation
9 Sound and acoustics
10 Graphics, effects and night vision
11 Throwables and equipment
12 Enemy AI
13 Game modes
14 Armory, front end and settings
15 Maps and scenes
16 Architecture
17 Content pipeline and tests
18 Known issues and workarounds
19 Recommended next steps

1 Status at a glance

Area	What exists
Scenes	Armory hub, Test Range, Tactical Raid, Search & Destroy, Hardpoint — deployed from the Armory
Weapons	5.56 carbine, .300 WM bolt-action, 7.62 battle rifle, 9 mm sidearm, all with moving mechanisms
Attachments	14 parts across 7 slots, live-swappable, with real mass and balance effects
Ammo, equipment	5.56, 7.62, 9 mm, .300 WM, fragments (G1/G7 drag) · M67 frag, M84 stun, M18 smoke, C4
Movement	Stances, lean, mounting, slide and slide cancel, dive, jump, vault, mantle, physical doors
AI	Squads with CQB room clearing and door use, three voices, four difficulty levels
Game modes	Tactical Raid / Extraction, Search & Destroy, Hardpoint
Graphics	Physical sky, SSGI, SSR, contact shadows, weathered materials, combat VFX, night vision
Audio	48 kHz synthesized gunfire, four room tails, supersonic cracks, gear foley, footsteps, master limiter
Front end	Deployment table, loading screen, key rebinding, hit markers, damage arcs, death camera
Persistence	JSON profile with weapon blueprints, loadout, settings and key bindings
Tests	165 PlayMode tests (160 pass, 5 opt-in render captures skipped) plus visual captures on demand

Milestones (all complete)

#	Milestone	Commit
M1	Project init, assemblies, InputReader, EventBus, pooling, setup wizard	`80d1fba`
M2	Locomotion biomechanics, stances, leaning, mounting, breathing	`38dafd5`
M3	Springs, weapon inertia, barrel collision, procedural ADS, PiP optics, recoil	`7482bb4`
M4	Weapon/caliber data, model rig and sockets, chamber tracking, staged reloads	`ce375b7`
M5	Ballistics sim, penetration/ricochet, shell casings, health and armour	`bb38a9b`
M6	Acoustic propagation, muzzle VFX, diegetic HUD	`2d5db22`
M7	Gunsmith and attachment physics	`93c506a`
M8	Tactical AI and squad CQB	`65a13f5`
M9	Game modes, match director, tactical equipment	`8367d25`
M10	3D gunsmith workbench, loadout persistence, HDRP settings menu	`cc73f52`
Fix	Editor workaround for an HDRP 17.6 volumetric fog bug	`1b05b0a`
M11	Meshy asset pipeline, procedural models, weapon mechanisms, battle rifle and sidearm	`b47e5f8`
M12	Photoreal HDRP (SSGI/SSR, physical sky), weathered materials, combat VFX, night vision	`9ad0a61`
M13	Physical gunshot synthesis, environment tails, supersonic cracks, gear foley	`0fddccf`
M14	Vault and mantle, slide cancel, compressed ready, physical doors, inspect, camera kick	`ce25ec5`
M15	Deployment table, scene flow, loading screen, complete settings and key rebinding	`f7ea4b8`
M16	Hit confirmation, damage attribution, Windows player builder	`5e82f20`
Fix	Playtest: battle rifle irons, station triggers, HUD under menus, AI NavMesh, audio mix	`957a772`
Audio	Footsteps for player and soldiers, click-free regenerated clips, trimmed voices	`1b8da03`
Polish	Damage direction, death camera, weapon racks, view poses, scope radiance	`da49e6e`

Each was verified with batch builds, the full PlayMode suite and render captures, then pushed.

2 Getting started

Play the built game

Run `Desktop\Claude of duty Build\ClaudeOfDuty.exe`. It opens fullscreen in the Armory hub. Walk to the **deployment table** and press F: pick an operation and an AI difficulty, then Deploy, and a loading screen takes you there. Escape opens the options menu with **Return to Armory** (outside the hub) and **Quit to Desktop**.

Open the project

- Unity Hub → Add → project from disk → the repository folder, editor **6000.6.4f1**.
- The builders below are already applied and committed; they are idempotent, so rerun any of them after a settings reset.
- Active Input Handling is *Input System Package (New)*.
- To play in the editor: open `Assets/_Project/Scenes/Armory.unity` (or any scene there), press Play, then **click inside the Game view** so it takes keyboard and mouse focus.

Editor builders (`Tools/FPS Builder`, run in order)

#	Menu	Creates
1	Initialize Project Layers & Tags	Linear colour space, HDRP asset, layers, collision matrix
2	Create Core Assets	`Data/Core/InputReader.asset` wired to the input actions
3	Create Test Weapons	Calibers; carbine, marksman, battle rifle and sidearm prefabs with mechanisms, data, recoil profiles and magazines; all gunsmith parts
4	Create Player Prefab	Rig, viewmodel, IK arms with sleeves and gloves, loadout, equipment, traversal, night vision, hit markers, profile manager, settings menu
5	Build Test Range	Locomotion Test Range, AI soldier prefabs, grenades, voices, Squad Alpha, CQB shoot house with doors, weapon racks, NavMesh
6	Build Weapon From Selected Model	Wraps the FBX/GLTF selected in the Project window in a socketed weapon prefab with data
7	Build Game Mode Scenes	Mode_Raid, Mode_SearchAndDestroy, Mode_Hardpoint (arena, doors, objectives, spawns, NavMesh, map schematics)
8	Build Armory	Armory hub with deployment table and weapon racks, catalog, HDRP dynamic resolution; every deployable scene in the build settings
9	Build Windows Player	`..\Claude of duty Build\ClaudeOfDuty.exe` (next to the repository, never inside it) from the build settings scenes

Command line

```
REM Full PlayMode test suite (headless)
Unity.exe -batchmode -nographics -projectPath . -runTests -testPlatform PlayMode -testResults results.xml

REM Visual check: renders PNGs of every scene (needs graphics, so no -nographics)
set FPS_CAPTURE_DIR=C:\captures
Unity.exe -batchmode -projectPath . -runTests -testPlatform PlayMode -testFilter FPS.Tests.RenderCapture

REM Windows player build (builder 9; exits non-zero on failure)
Unity.exe -batchmode -quit -projectPath . -executeMethod FPS.EditorTools.PlayerBuildTool.BuildWindowsPlayer
```

> Close any editor that has this project open before running batch commands — Unity locks the project.

3 Controls

Action	Keyboard & mouse	Gamepad
Move / look	WASD / mouse	Left / right stick
Fire	Left mouse	Right trigger
Aim down sights	Right mouse	Left trigger
Sprint (hold) · tactical sprint (double-tap)	Left Shift	L3
Hold breath (while aiming)	Left Shift	L3
Walk slowly	Left Alt	—
Walking pace up / down · crack a door (while holding Interact)	Mouse wheel	—
Jump · vault or mantle what is ahead · cancel a slide	Space	A (tap)
Crouch (tap) · slide when sprinting · cancel a slide	C	B (tap)
Prone · dive when sprinting	Hold C, or Left Ctrl	Hold B
Lean left / right	Q / E	D-pad ← / →
Reload (tap) · speed reload (second tap) · magazine check (hold)	R	X
Change fire mode	B	D-pad ↓ (tap)
Inspect weapon	I	D-pad ↓ (hold)
Switch to canted sight	X	D-pad ↑
Mount weapon on ledge or corner · bash a door you face	V	R3
Primary / secondary weapon	1 / 2	Y cycles
Interact (tap) · hold-to-use objectives and door cracking (hold)	F	A (hold)
Weapon light and laser on/off	T	View (tap)
Night vision on/off	N	View (hold 0.4 s)
Throw lethal grenade (hold to cook)	G, mouse forward	RB
Throw tactical grenade (hold to cook)	H, mouse back	LB
Options menu · leave the workbench or deployment table	Escape	Start

Tap/hold threshold is 0.35 s. On the gamepad A is shared: a tap jumps or climbs, a hold interacts. The menu key lives in its own *System* input map that stays on while gameplay input is paused, so menus and stations always respond. All input goes through one `InputReader` asset: continuous state is polled, discrete intents are events.

Rebinding: Options → Controls rebinds 22 keyboard bindings (Escape cancels). Keys shared by a tap/hold pair move together — Reload with magazine check, Crouch with Prone, Sprint with tactical sprint and hold breath. Overrides are saved in the profile.

4 Movement, traversal and the player's body

Locomotion

- **Paces** (m/s): slow walk 1.3, combat pace 3.4, sprint 5.8, tactical sprint 7.2. The mouse wheel trims walking pace in 10 % steps down to 30 %.
- **Modifiers**: crouched 55 %, prone 22 %, aiming 60 %, backwards 65 %, strafing 80 %.
- **Momentum**: acceleration 8 m/s^2, deceleration 10 m/s^2; less control while sprinting and almost none in the air.
- **Traction** comes from the ground's physics material (dynamic friction 0.6 = full grip). Ice makes you slide and overshoot.
- **Weapon load**: every kilogram carried costs 1.2 % speed (floor 82 %); ergonomics affect handling.
- **Stamina**: sprint drains 7 %/s, tactical sprint 20 %/s; regenerates 12 %/s after 1 s. Running dry exhausts you until 35 % is back.
- **Slide**: crouch while sprinting above 4.5 m/s for an 8 % boost; slope physics and friction apply, steerable, up to 2.5 s. **Slide cancel**: crouch or jump mid-slide to end it in a crouch, with the weapon ready 0.15 s later instead of after the sprint-to-fire delay.
- **Dive**: go prone while sprinting to dive forward (0.55 s) and land prone.
- **Injuries**: leg damage cuts sprint speed by up to 45 % and blocks tactical sprint past 50 % injury.

Traversal: jump, vault and mantle

- Press Space (A) while moving into an obstacle, or hold it in the air. The controller probes the face (0.35–2.05 m), the top, its depth, the ground behind and capsule clearance.
- **Vault**: obstacles 0.5–1.1 m tall and at most 1 m deep with ground behind are crossed in one motion.
- **Mantle**: anything else up to 2.1 m with room to stand is climbed onto (ending crouched if standing does not fit). Roofed or taller ledges cannot be climbed.
- The path rises before it advances. The camera pitches and rolls with the climb, the weapon swings to a one-handed side carry (no firing) and the support hand plants on the edge.
- **Jump**: a 3 m/s take-off when nothing is climbable; from a crouch you stand up first.

Stance, lean and mounting

Stance	Capsule height	Eye height	Transition
Standing	1.8 m	1.66 m	stand ↔ crouch 0.28 s
Crouched	1.15 m	1.02 m	crouch ↔ prone 0.65 s
Prone	0.4 m	0.28 m	stand ↔ prone 0.9 s

- Clearance checks stop you standing up under a low roof. Prone aligns the body to slopes up to 30°, limits pitch to −30°/+25° and cannot swing through walls.
- **Lean**: ±18° about the hips, limited by sphere-casts so you never lean through a wall.
- **Mounting** (V): rest the weapon on a ledge or wall corner within reach. Sway drops to 20 % and recoil to 55 % inside a limited arc (ledges 40°×18°, corners 25°×25°).

Doors

- Physical doors (shoot house, warehouse) swing both ways, always away from the user. Tap F to open or close.
- **Crack**: hold F and scroll to open the door 6° a step, up to 35°, to peek through the gap.
- **Bash**: V facing a door, or sprinting into it, slams it open and stuns soldiers in its sweep. A gentle swing stops against bodies.
- The leaf is wood, so rounds go through it. Soldiers push closed doors open as they walk through.

Physiology

- **Heart rate** 65–185 bpm. Exertion drives it (sprint 142, tactical sprint 172) and suppression adds stress; rises over ~3.5 s, recovers over ~14 s.

- **Breathing** (12–42 breaths/min) drives a figure-8 weapon sway of 0.18°–1.4°, plus slow wander, a heartbeat twitch and a fast **hand tremor** (three noise octaves from 9 Hz, 0.035° scaled by heart rate; less when mounted, more with arm injuries). Crouching reduces sway to 75 %, prone to 45 %.
- **Hold breath** (Shift while aiming) steadies sway to 12 % for 2.5–7 s depending on heart rate; running out costs a gasp with heavy sway.
- **Headbob** is gait-synchronised per pace, with a landing dip proportional to fall speed, a slide lower-and-roll and a dive pitch.

Health and feedback

- 100 HP. Hit zones (head, thorax, stomach, arms, legs) follow stance, lean and prone, so enemy fire hits what is exposed.
- **Suppression**: supersonic rounds within 2 m of your head flinch it away from the crack, blur the periphery and raise heart rate.
- **Stun whiteout**: on-screen flashes blind fully, off-screen ones fade with angle; deafness muffles the mix and a tinnitus ring fades in and out.
- **Stress**: heavy damage and suppression close a vignette and drain colour.
- **Damage direction**: each hit draws a thin dark-red arc round the screen centre toward its source (the shooter's muzzle or the blast): full for 0.2 s, gone by 1.5 s, one per attacker, four at most. Switch it off in Options → Gameplay.
- **Death camera**: the rifle disappears and the view drops to the floor over 0.8 s, rolling onto its side; control returns only on a respawn.
- **Interaction**: look at an interactable within 2.2 m and press F (crates, locker, racks, workbench, deployment table, doors, fallen soldiers' rifles, objectives); hold F for hold-to-use objectives with a HUD progress bar.

5 Weapon handling, aiming and recoil

The first-person weapon hangs from a chain of pivots, each written by exactly one system: **Inertia** → **Recoil** → **Action poses** → **Aim** → **weapon model**, with two-bone IK arms. Every rotation acts about the point where the stock meets the shoulder.

Inertia and physical presence

- **Spring handling**: the weapon follows the view through damped springs whose stiffness comes from mass, ergonomics and the build's measured **rotational inertia** about the shoulder. Long barrels and suppressors swing visibly slower.
- **Free aim**: quick turns lead inside an elastic deadzone (2.5°×4° at the hip, 25 % when aiming, 10 % mounted).
- **Movement**: trail and strafe roll, sprint posture, and a high-ready posture for tactical sprint.
- **Barrel collision**: a sphere-cast along the barrel first slides the weapon back into a **compressed ready** (stock under the arm: it still fires, but cannot aim), then tilts it to a high ready (short barrels) or low ready (16"+). Past a quarter tilt the weapon is tucked and can neither fire nor aim.
- **View poses**: the high and low ready, the climbing carry and the magazine check are held in the view, so the rifle stays on screen whatever its length.

Aiming and optics

- **Procedural ADS**: no animation. Each frame the pose is solved that places the active sight's aligner on the eye line at its eye relief. Base ADS time 0.3 s, scaled by ergonomics and parts.
- **Canted sights**: X rolls the weapon 45° onto offset irons.
- **Picture-in-picture scopes**: a second camera renders into the lens with FOV = lens angular size ÷ magnification, so the image is optically consistent. The scope renders scene radiance and the main camera exposes it once, so the picture is as bright as the world. The main view blurs in the periphery; look sensitivity scales with zoom.
- **Scope lens shader**: the PiP image passes through barrel distortion, rim chromatic aberration, an eye-relief shadow disc and a field stop, with an etched duplex-dot or chevron reticle and an HDR lit centre.
- **Night-vision active aim**: with goggles down and a laser fitted, aiming cants the weapon 45° with the IR laser forced on (no optic, no zoom).
- Red dots and irons are unmagnified and cost no extra camera.

Recoil and camera kick

- Per shot: muzzle rise, biased sideways torque and roll; rearward shoulder push and a small upward kick; a delayed **bolt-carrier jolt** ~35 ms later; **permanent view climb** you pull down yourself; a damped camera concussion with no random shake.
- Ramps up during sustained fire and resets after a 0.3 s pause. Aiming 75 %, mounted 55 %. Scales with **momentum** — heavier builds kick less — and per-axis attachment multipliers.
- **Camera kick**: each unsuppressed shot of your own adds a 1.5° field-of-view punch scaled by √(muzzle energy ÷ 1700 J) (0.4–1.8×) over 40 ms, with a chromatic-aberration pulse.

Inspect

I (D-pad ↓ hold) plays a 3.5 s keyframed inspection: the left side and markings, then the ejection port. Aiming, sprinting, climbing, reloading, a magazine check or firing cancels it.

Arms

Two-bone IK keeps the firing hand on the pistol grip. The support hand moves between the handguard, a fitted foregrip, the magazine (reloads and checks), the charging handle (cycling) and the ledge during a climb. Sleeves and gloves are modelled in normal-mapped woven cloth and pored leather.

6 Weapon mechanics and weapons

- **Data split**: `WeaponDataSO` holds the firearm (mass, ergonomics, barrel length, action, fire modes, RPM, MoA, capacity, stage timings, recoil and audio profiles); `CaliberDataSO` holds the ammunition (mass, diameter, velocity vs barrel length, drag model and BC, penetration, damage).
- **Chamber tracking**: closed-bolt and bolt-action weapons start at capacity + 1; open bolt never carries a +1. Hold-open locks the bolt back on the last round; bolt actions cycle after the trigger is released.
- **Fire modes**: semi, 3-round burst, full auto at the cyclic rate.
- **Firing**: every round leaves the **muzzle socket along the bore** with MoA dispersion — never from screen centre. Muzzle velocity comes from caliber and actual barrel length. Fire cancels sprint after a sprint-to-fire delay; tucked barrels, climbs, dives and prone transitions block it.
- **Staged reloads** (magazine out → in → charge only if empty; bolt release 0.6× when locked back): tap R = tactical (partial magazine back in the pouch); second tap during removal = speed reload (physical magazine drops); hold R = magazine check (the magazine comes into view; estimate from "empty" to "full"). Sprint, dive, swap or firing the chambered round interrupts; the next press resumes after the last completed stage.
- **Mechanisms**: trigger, selector, bolt carrier or slide (cycles every shot, held back when locked open), charging handle (charge stage, or rides the carrier) and bolt handle move on the model, for the player and for AI soldiers. Bolt actions come off the sights for the bolt stroke; the battle rifle's flip-up irons fold flat under an optic.
- **Swaps**: holster/draw times scale with ergonomics; each weapon keeps its rounds, chamber and reload progress while holstered.
- **Loadout**: carbine primary, marksman secondary. **Weapon racks** (Armory and test range) swap the primary to the battle rifle and the secondary to the sidearm, and back, saving the loadout. Racks show their weapons; the carried one is missing.
- **Casings**: pooled physical casings eject with spin and clink per surface. **Resupply** at ammo crates; **looting** a fallen soldier's rifle takes its loaded magazines (up to loadout + 2).

Weapon	Action	Build and notes
Test Carbine 5.56	Closed bolt, semi/auto, 800 RPM, 2.5 MoA, 30 rounds, 4 spare magazines	Modular receiver with barrel, muzzle, optic, underbarrel, stock, laser and light mounts; 45° canted irons. Default build ≈ 3.47 kg: 14.5" barrel, flash hider, 4× prism scope, angled grip, fixed stock, light, laser; ≈ 899 m/s.
Test Marksman .300 WM	Bolt action, 60 RPM, 0.6 MoA, 5 rounds	6.2 kg, fixed 8× PiP scope, 24" barrel, heavy recoil, manual cycling.
Battle Rifle 7.62	Closed bolt, semi/auto, 600 RPM, 1.5 MoA, 20 rounds, bolt hold-open, 4 spare magazines	16" barrel; optic, underbarrel, laser and light mounts. Default build ≈ 3.84 kg: red dot, vertical foregrip, weapon light. 7.62 hits harder than the carbine's 5.56; ADS 0.36 s.
Sidearm 9 mm	Closed bolt with reciprocating slide, semi, 4.5 MoA, 15 rounds, slide locks back on empty, 3 spare magazines	0.85 kg, 4" barrel, fixed sights. Quick handling: ADS 0.18 s, sprint-to-fire 0.1 s, holster 0.3 s, draw 0.35 s.

7 Gunsmith and attachments

- **Slots**: barrel, muzzle, optic, underbarrel, stock, laser, light. Each part attaches at a `{Slot}Mount` anywhere under the weapon, so a barrel carries the muzzle mount. The carbine takes every slot; the battle rifle takes optic, underbarrel, laser and light; the marksman rifle and sidearm are fixed.
- **Sockets move with parts**: a suppressor becomes the new muzzle (rounds and flash leave its end), a foregrip moves the support hand, a collapsed stock moves the shoulder point.
- **Live stats**: mass, ergonomics, barrel length → muzzle velocity, recoil multipliers, flash/gas size, ADS time, suppressed audio profile, and the **measured centre of mass and moment of inertia** about the shoulder (from each part's real position), which drives the handling springs.
- Builds are applied to a runtime clone of the weapon data — assets are never modified. Live refits keep rounds, chamber and reload progress.

Part	Mass	Effects
10.5" CQB barrel	0.45 kg	+6 ergonomics, flash ×1.5, ADS ×0.94, lower velocity
14.5" carbine barrel	0.60 kg	Baseline
20" marksman barrel	0.85 kg	−8 ergonomics, flash ×0.8, ADS ×1.08, full 940 m/s
Birdcage flash hider	0.08 kg	Flash ×0.25
Three-port compensator	0.12 kg	Vertical recoil ×0.8, horizontal ×0.85, flash ×1.4
5.56 suppressor	0.55 kg	No flash, suppressed sound, vertical ×0.9, rearward ×0.85, +1 % velocity, −8 ergonomics, front-heavy
4× prism scope	0.50 kg	PiP magnified optic, −4 ergonomics, ADS ×1.1
Red dot sight	0.15 kg	Unmagnified, ADS ×0.95
Vertical foregrip	0.12 kg	Horizontal recoil ×0.85, +4 ergonomics, vertical hand pose
Angled foregrip	0.08 kg	Vertical recoil ×0.92, ADS ×0.95, +6 ergonomics
Fixed stock	0.40 kg	Rearward recoil ×0.9
Collapsed stock	0.30 kg	+4 ergonomics, ADS ×0.95, rearward ×1.1, sits closer
Visible laser	0.09 kg	Beam and dot on whatever it hits; infrared beam under night vision
Weapon light	0.12 kg	15,000 cd spot, 40° spill; lights the volumetric fog; makes the carrier visible to AI

8 Ballistics, damage and hit confirmation

- **Simulation**: one projectile per shot with gravity and **G1/G7 drag** against the wind, sub-stepped at 1/120 s and resolved with one batched `RaycastCommand` per frame (no per-shot allocation). Up to 4 interactions per round per frame; rounds expire after 6 s or below 80 m/s.
- **Penetration**: energy cost per metre = surface resistance (0.007 × density × hardness) × bullet cross-section (mm^2) × an AP factor (1.5 → 0.5 by penetration power). Thickness is measured by casting back from the far side; exit speed comes from leftover energy, with deflection. Drywall is shot through easily; 30 cm of concrete stops rifle rounds.
- **Ricochet**: impacts shallower than the surface's ricochet angle glance off with probability 1 − angle/ricochetAngle.
- **Near misses**: supersonic rounds within 2 m of a listener produce a crack and suppression.

Calibers

Caliber	Bullet	Velocity	Pen	Dmg
5.56×45 M855	4.0 g, G7 .151	940 m/s @20"	40	35
7.62×51 M80	9.5 g, G7 .200	840 m/s @22"	55	55
9×19 124 gr	8.0 g, G1 .150	360 m/s @4"	15	30
.300 Win Mag	12.3 g, G7 .290	880 m/s @24"	70	90
Fragment	0.5 g, G1 .012	1300 m/s	12	22

5.56 loses 7.5 m/s per inch of barrel below 20".

Surfaces

Surface	Density kg/m^3	Hardness	Ricochet
Concrete	2400	60	20°
Brick	1900	45	15°
Wood	600	35	8°
Drywall	700	3	3°
Metal	7850	100	30°
Glass	2500	12	5°
Dirt	1600	35	6°
Flesh	1050	15	—

Bodies and armour

- **Health**: 100 HP. Zone multipliers: head ×3.5, thorax ×1.4, stomach ×1.1, arms ×0.55, legs ×0.7; limb capacity 40. Rounds can pass through flesh and continue.
- **Armour**: Level IV ceramic front and back plates (rating 60, durability 80, 3.2 kg each) and a Level IIIA helmet (rating 25, durability 40, 1.4 kg). Coverage depends on direction — face shots get under the helmet.
- **Defeat roll**: a logistic roll of energy-scaled penetration against protection (worn down by hits) decides stop (blunt trauma) or defeat (residual absorption).
- **Injuries**: legs slow sprinting; arms add sway.

Damage attribution and hit confirmation

- **Attribution**: every round carries its attacker into the hit. Grenade fragments and blasts credit the thrower, so every damage event names who caused it.
- **Hit markers**: each hit you cause (never on yourself) pops a small X at the screen centre — white for flesh, pale blue when armour stopped the round, red for a kill, larger red for a head-shot kill. It shows for 0.22 s, or 0.45 s on a kill.
- **Hit ticks**: a 2D sound per hit — a soft mid tick for flesh, a brittle ceramic crack for armour, a deep double tick for a kill, the kill plus a bright ring for a head shot. Override them under `Art/Audio/Overrides/HitMarker/`.
- **Setting**: Options → Gameplay → hit markers Off, Audio only, Visual only or Audio and visual (default).
- **Damage direction** on the player's side: see section 4.

9 Sound and acoustics

Physical gunshot synthesis

- Every clip is synthesized by the editor (`AudioSynth`, 48 kHz 16-bit) from physical models: a Friedlander blast wave, asymmetric air overload (the rarefaction clips harder than the shock), a phase-aligned sub-bass sweep, a dispersive coil-spring twang, clicks, and room impulse responses (reflections, flutter, darkening diffuse decay, rumble) convolved with the dry blast.
- **Layers per weapon**: transient (200 Hz–8 kHz, 3 variants), sub-bass (30–90 Hz), mechanism, body, distant report and a low-ammo ping. Mechanisms differ by action: direct impingement (extractor snap, buffer impact and spring twang, carrier slam ~55 ms, trigger reset), piston, pistol slide and bolt action.
- Suppressor and explosion profiles (frag, stun, C4) use the same layers. A suppressed rifle is quieter and duller and carries 300 m instead of 1500 m.

Propagation and rooms

- **Own shots**: blast, sub and body on one voice, the action on its own. **Other shots** arrive after **distance ÷ 343 m/s** as a close report (with sub) or, beyond the distant threshold, a distant report.
- **Low ammo**: below 3 rounds the magazine pings and the action rises in pitch.
- **Reverb tails**: a 9-ray probe around the listener weighs four stereo tails; indoor weight splits between small room and warehouse by hall size (1.6 × ceiling height or mean wall distance, 5 → 12 m).

Tail	Length	Character
Small room	0.4 s	Tight, bright slap
Warehouse	1.4 s	Long hall ring
Urban canyon	2.2 s	87 ms flutter echo between façades
Open field	3 s	Rolling low thunder

Footsteps

- A step per foot strike for the player (at the bottom of each head bob) and every soldier (from their gait). The clip comes from the surface underfoot: steps, prone crawl and landings.
- Loudness follows the noise radius the AI's hearing uses (sprint loud, crouch-walk quiet); landings scale with fall speed. Own steps play in the head, others' travel at 343 m/s within their radius.

Supersonic cracks

Each supersonic near miss solves the Mach cone: the crack comes from upstream of the closest point and arrives later than the bullet passed; its N-wave length (pitch) and overpressure (volume) follow the Mach number and miss distance, and a receding whip follows. Cracks come from ahead of the shooter — the shooter is behind the sound.

Foley and feedback

- **Gear**: a plate-carrier rustle that follows body acceleration, turn rate and speed; sling-swivel clinks on weapon swings and hard landings.
- **Handling**: dry fire, magazine out/in, bolt, fire-mode switch, magazine check, for you and for nearby soldiers. **Impacts** and casing clinks per surface, delayed by distance.
- **Muffling**: suppression, low health and stun deafness sweep a low-pass filter on the listener; flashbangs add a fading tinnitus ring.
- **Volumes**: master, effects and voice sliders in Options → Audio.
- **Mix**: each shot's tail rings on its own voice (four per space), so reverb builds in sustained fire; own shots keep headroom, and a peak limiter rides overs down instead of clipping. Every clip ends with a short fade; voices are trimmed so callouts start at once.

Master-bus dynamic range (Options → Audio)

Preset	Compressor	Makeup	Limiter	Character
War Tapes	−10 dB, 1.5:1	+4 dB	Soft, −0.5 dBFS	Loud and wide, saturating
Studio Reference	Transparent	0 dB	−0.3 dBFS ceiling	Uncoloured
Night Mode	−30 dB, 5:1	+12 dB	−6 dBFS ceiling	Lifts footsteps, tames gunfire

> **Replacing sounds.** Drop a WAV at `Art/Audio/Overrides/<Owner>/<Layer>.wav` (or `<Layer>_*.wav` for variants) — owner is the weapon name, `<Weapon>_Suppressed`, `Grenade_Frag`, `Bullet`, `Gear`, `HitMarker` — and it replaces that synthesized layer on the next build. Soldier callouts use Windows speech synthesis in three voices; assign recorded lines in the voice profiles.

10 Graphics, effects and night vision

Pipeline and world look (HDRP 17.6)

Feature	Setting
Colour space / buffer	Linear; HDR colour buffer; deferred and forward lit
Sky and sun	Physically based sky with a generated cumulus cloud layer that casts shadows; sun 90,000 lux at 5,600 K
Exposure / tonemapping	Histogram auto exposure EV 4.5–15.5; ACES
Shadows	4096 atlases, 32-bit directional depth, up to 128 requests; contact and micro shadows
Reflections / GI	SSR, SSGI (half resolution), SSAO; box-projected reflection probe per interior; ray tracing off
Volumetrics	Thin global volumetric fog plus local fog volumes (smoke, dust), up to 256 on screen
Lens	Film grain 0.18, chromatic aberration 0.08, split toning
Decals / distortion / motion	Decals (bullet holes, thermal rings), distortion (heat haze), motion vectors (motion blur, TAA)
Dynamic resolution	Hardware scaling 50–100 %, TAA upscale, DLSS and FSR2 when the platform reports them

- **Interiors**: every room (shoot house, warehouse rooms, workshop, indoor range) gets a reflection probe and a local volume that cuts the open-sky ambient its roof blocks.
- **Lights**: interior lamps are physical point lights in lumens; weapon lights are spot lights in candela that light the fog. AI sight uses the same light.

Weathered materials

- Level geometry gets a weathered HDRP LayeredLit material by ballistic surface: the surface, a grime layer and, on floors and ground only, wet patches — all world-space triplanar with height-based blending.
- Procedural models use generated finishes (base, normal and mask maps per finish); worn finishes show bare metal on chamfered edges.

Combat effects

- **Muzzle**: a one-frame point-light flash (40,000 lm) and gas burst, a shock cone (~45 ms, scaled by √flash), HDR embers and, at most every 0.12 s, a volumetric smoke wisp. Suppressed builds skip the flash, shock and embers. Barrel heat drives heat haze and lingering breech smoke.
- **Impacts**: pooled decals sized by caliber plus surface-tinted dust. Steel throws colliding sparks along the reflected path (double on a ricochet) and leaves a glowing thermal ring that fades over 3 s; masonry and soil leave a lingering volumetric dust cloud.
- **Particles** use HDRP/Unlit with a 4×4 alpha flipbook, tinted per instance. Smoke grenades are a local volumetric fog volume plus billows.

Night vision (N)

- The goggles flip a volume in over 0.35 s: white (P45) or green phosphor, auto-brightness exposure down to EV −4, bloom halos, scintillation grain and a round tube vignette.
- Lasers switch to infrared beams that only show through the goggles; with a laser fitted, aiming becomes canted active aim. Phosphor colour is in Options → Gameplay.

Cameras, post-processing and HUD

- **Main camera** renders the world and the first-person weapon in-world, so the weapon receives real lighting and shadows. The **PiP scope camera** renders to an `R16G16B16A16_SFloat` target; volumes are split by layer (scope-only: vignette, no tonemapping, no motion blur; main-only: peripheral depth of field while scoped).

- **Feedback volumes**: suppression focal blur, stress vignette and desaturation, stun whiteout, shot-impulse chromatic aberration, motion blur setting.
- **HUD** (built in code): compass tape with bearing, interaction prompts, fire-mode and magazine status only on a mode switch or magazine check, hit markers, no crosshair by default. Match HUD: mode, objective, clock, score, round banners, hold-progress bar.

11 Throwables and equipment

- **Throwing**: press lowers the rifle and pulls the pin (0.35 s); the spoon flies and the fuse burns while held (**cooking**). Release throws along the view with 10° loft plus your own velocity, keeping the fuse already used. Hold too long and it goes off in hand; dying or being disabled mid-cook drops it live at your feet.
- **Default loadout**: 2 frags (lethal) and 2 stun grenades (tactical). The equipment locker swaps stun ↔ smoke; ammo crates restock grenades.

Item	Fuse	Effect
M67 frag	4.5 s	0.4 kg. **64 real fragment projectiles** through the ballistics sim (they penetrate, ricochet and suppress); blast overpressure within 4 m, blocked by solid cover; concussive push on casings, magazines, rifles and bodies.
M84 stun	1.5 s	Blinds fully within 4 m, fading to 18 m, by facing and line of sight; deafens to 25 m. Through a wall: 15 % blindness, 60 % deafness.
M18 smoke	1.5 s	≈ 4.5 m cloud: builds 3 s, holds 35 s, fades 6 s. Local volumetric fog + particles, and an obstruction AI cannot see through (thin edges partly).
C4 charge	40 s	Search & Destroy objective: 10 m blast, 48 fragments, large report, quickening beep.

12 Enemy AI

The soldier

- Procedural body on a NavMesh agent: crouch/stand heights, ±22° lean from the hips, body turns to follow the aim, shoulder aim pivot with turn-rate limit and recoil, head look, IK arms on the rifle.
- Modelled operator gear — friendlies in coyote/tan, hostiles in ranger green/olive — over invisible primitive hitboxes on the bones.
- Carries the same gunsmith-built carbine as the player (red dot, light, flash hider) with identical ballistics, audio and moving mechanism; rounds leave its muzzle along its bore; magazines and chamber are counted; reloads use real timings and foley.
- Kit: one frag, one stun grenade, a helmet, one of three voices. On death the body collapses physically and drops a lootable rifle. Soldiers push closed doors open; a bashed door stuns them.

Perception (≈ 10 Hz)

- **Sight**: awareness builds per target from visible body fraction (head, chest, pelvis lines), **light level** (dark corners shrink sight range to 12 %), stance (crouched 65 %, prone 35 %), motion, distance (90 m), and focus vs peripheral vision (30° / 80°, peripheral at 35 % speed). Under 4 m it is three times faster. Full awareness = identified ("Contact!"), remembered for 25 s.
- **Smoke** thins or blocks sight lines; muzzle flashes still glow through thin smoke and give an unsuppressed shooter away instantly.
- **Hearing** alerts but never identifies: footsteps by speed and stance (halved through walls); gunshots to the weapon's audible range, with suppressed shots located far worse (30 % vs 8 % error).
- **Stress**: near misses and hits raise suppression and turn the soldier toward the shooter; stun grenades blind and deafen.

Tactics

- **Idle / suspicious**: hold a post or patrol; look toward noises. **Alerted**: move up on the last known position, pre-aiming around corners (slicing the pie).
- **Firefight**: NavMesh cover search against the threat — low cover (crouch to hide, stand to fire) or tall cover (lean out) — avoiding teammates' claims; peek-and-hide cycles; 0.35–0.7 s reaction; aim error grows with own suppression, target speed and moving fire, and settles while tracking; bursts of 3–6; never fires into its own cover or through a teammate.
- **Lost contact**: suppressive fire on the last known position for 6 s, then hunt. A target holding still behind cover for 4 s gets a **cooked frag** (6–28 m, arc solved to clear the cover).
- **Reloading**: callout, duck into cover, a teammate covers.

Squads

- **Shared knowledge**: one sighting engages the whole squad.
- **Suppress and flank**: when the target goes to ground, one soldier flanks ("Flanking left/right!") while the rest suppress; "Target suppressed!" when squad rounds crack past the target's head.
- **CQB room clearing**: stack on the nearest door, flashbang in while the stack looks away, then one man at a time slices the pie through the doorway to his point of domination; "Clear!" if empty.
- **Coordination**: grenade cooldowns, cover claims, team-wide friendly-fire checks (player included), "Covering!", "Man down!".
- **Objectives**: hold a point (spread around it) or work it (plant/defuse while others cover); a dead worker is replaced; threats are not chased beyond a 25 m leash.
- **Callouts**: Contact, Reloading, Covering, Flanking left/right, Target suppressed, Frag out, Flashbang out, Stack up, Go go go, Clear, Lost contact, Man down.

Difficulty

Level	Awareness gain	Sight range	Aim spread	Reaction time
Recruit	×0.6	×0.8	×1.8	×1.6
Regular (authored)	×1	×1	×1	×1
Hardened	×1.3	×1.1	×0.75	×0.8
Realism	×1.6	×1.2	×0.55	×0.65

Chosen at the deployment table or in Options → Gameplay and saved in the profile. Values are read at use, so a change applies at once.

13 Game modes

Every mode is deployed from the Armory's deployment table. All modes run on a **match director**: the player plus AI soldiers per team in squads (friendlies tan, hostiles olive); phases pre-round countdown → live → round over → match over; clock, round wins and scores; respawns after the mode's delay (or one life) with the player's controls removed while dead; fresh soldier bodies each spawn and corpse clean-up.

Smart spawns score every point: −60 if a living enemy can see a body standing there; distance to the nearest enemy (capped at 40 m) and −100 inside 12 m; +4 per nearby teammate (up to 3); a penalty for points used in the last 5 s.

Mode	Rules	Teams
Tactical Raid / Extraction	One life, no resupply. You start behind concrete T-walls, out of the garrison's sight. Complete the objectives by holding Interact (download intel 6 s, plant tracker 5 s); each completion raises the alarm across the garrison. Then reach the LZ and hold inside it for 8 s. Death or the 15-minute clock fails the mission. Ammunition is what you carry plus magazines looted from the dead.	Solo vs a 6-man garrison in two squads
Search & Destroy	120 s rounds, one life. Attackers plant C4 at site A (warehouse) or B (container yard): 5 s hold, 40 s fuse with a quickening beep. Defenders stop the plant or defuse (7 s hold). Elimination wins unless a planted charge still needs defusing; a planted charge stops the clock. Sides swap after round 3; first to 4 wins. AI attackers send a planter and cover him; defenders split over the sites; everyone converges on a planted charge.	Player + 2 vs 3
Hardpoint	One of three zones is live for 60 s, then moves. Each second a team alone in the zone scores; contested zones score nobody. First to 150, or most after 10 minutes. 4 s respawns away from enemy sight and outside the live zone; all squads are sent to the live zone.	Player + 3 vs 4

14 Armory, front end and settings

The Armory hub

A lit workshop with the **gunsmith workbench**, the **deployment table**, primary and secondary **weapon racks** on the west wall, an ammo crate and the equipment locker, opening straight onto a **30 m indoor range** (targets at 10, 20 and 28 m; an unarmoured dummy at 15 m and an armoured one at 25 m). Test-fire a build seconds after making it.

Deployment table

- Interact to step up: the camera moves over the table, where a hologram shows the selected map.
- The tactical screen lists the operations — Tactical Raid, Search & Destroy, Hardpoint, Locomotion Test Range — each with a briefing and map schematic, plus the AI difficulty. Deploy saves the difficulty and loads the scene; Back or Escape leaves.

Scene flow and loading screen

- Scenes load asynchronously behind a loading screen: operation name, map schematic, progress bar and rotating field notes (gameplay tips). It stays up at least 1.5 s and fades out after the new scene's first frames.
- The options menu offers **Resume**, **Return to Armory** (outside the hub) and **Quit to Desktop**.

Gunsmith workbench

- Interact to step up: the bench camera shows the carried weapon on a turntable with a marker on every mount.
- Drag to turn it; click a marker to list compatible parts from the catalog; click a part to fit it. The model, measured balance and stats panel (mass, ergonomics, barrel, muzzle velocity, recoil, ADS time, swing inertia, flash or suppressed) update instantly.
- "Fit to weapon" applies the build to your rifle and saves it; "Leave" or Escape hands control back.

Profile and saves

- `profile.json` in the persistent data folder: weapon blueprints (weapon and parts by asset id), loadout (a blueprint per slot plus grenade ids), all settings and key-binding overrides.
- Atomic writes (`.tmp` then swap) keep a `.bak` ; an unreadable file is moved to `.corrupt` and the backup or defaults load. The saved loadout is applied on every spawn.
- Catalog ids are asset names — renaming an asset orphans it in saved profiles (missing parts are skipped with a warning).

Options menu (Escape) — applied live, saved with the profile

Tab	Settings
Video	Field of view 60–100°; scope resolution 512 / 1024 / 2048 px; motion blur Off / Low / High; upscaling Native, TAAU, FSR2, DLSS (FSR2 and DLSS only where detected, otherwise TAAU); render scale 50–100 %
Audio	Master, effects and voice volume sliders; dynamic range War Tapes / Studio Reference / Night Mode
Controls	Look and ADS sensitivity sliders; rebinding of 22 keyboard bindings
Gameplay	Night-vision phosphor (white / green); AI difficulty; hit markers (Off / Audio / Visual / Both); damage direction

Sliders save when the menu closes. The menu pauses the player while open; at the workbench or deployment table, Escape leaves the station instead.

15 Maps and scenes

Scene	Contents
Armory hub, first in build	Workshop, gunsmith workbench, deployment table, weapon racks, ammo crate, equipment locker, 30 m indoor range with targets and armoured dummies.
Locomotion Test Range sandbox	Movement: 15° and 35° slide ramps, ice sheet, standing and crouched mounting ledges, corner wall, crouch (1.3 m) and crawl (0.6 m) tunnels, narrow barrel-tuck corridor, vault and mantle obstacles. Shooting: range lane with targets at 25/50/100/200 m; penetration wall row (drywall, wood door, glass, 1 cm and 3 cm steel, brick, concrete) with plain and armoured dummies. CQB and acoustics: dusty dark shoot house (a CQB room with physical doors), street canyon. Ammo crate, equipment locker, weapon racks, and a cover yard held by Squad Alpha (facing away until gunfire brings them).
Mode_Raid · Mode_SearchAndDestroy · Mode_Hardpoint	One arena: roofed two-room warehouse (both rooms CQB-clearable, with physical doors), walled compound to the west, container yard to the east, cover lines across the middle; each scene carries its mode's director, objectives, spawns and HUD.

Every scene has a baked NavMesh (door hinges excluded), the photoreal world look with interior probes, and the full player prefab. **Map schematics** for the deployment table and loading screen are rendered top-down from each scene's colliders (shaded by height, 10 m grid) with objectives in amber and spawns by team.

16 Architecture

Assemblies (references only point down the table)

Assembly	Namespace	Scope
FPS.Core	FPS.Core	Input and rebinding, EventBus, pooling, layers, spring and IK math, audio levels, shared interfaces and event structs
FPS.Ballistics	FPS.Ballistics	Calibers, projectile sim, penetration/ricochet, casings, hitboxes, health, armour, attribution
FPS.Audio	FPS.Audio	Acoustics and tails, supersonic cracks, weapon audio profiles, impact decals/dust/sparks/audio, particles, dynamic range
FPS.Player	FPS.Locomotion	Locomotion, vault/mantle, stance/lean/mount, breathing, suppression and flash feedback, night vision, camera chain
FPS.Weapons	FPS.Weapons	Weapon data, model rig and sockets, mechanisms, viewmodel, optics, recoil, controller, gunsmith, equipment, gear foley
FPS.AI	FPS.AI	Perception, soldier body and weapon, tactical brain, squads, CQB rooms, doors, difficulty
FPS.UI	FPS.UI	Diegetic HUD, hit markers
FPS.Gameplay	FPS.Gameplay	Match director, modes, objectives, spawns, match HUD, profiles, workbench, racks, deployment table, scene flow, settings
FPS.Editor	FPS.EditorTools	Builders 1–9, Meshy pipeline, procedural models, audio synthesis, world look, validators (editor only)

Upward communication uses the zero-allocation `EventBus<T>` (struct events: ShotFired, Gunshot, WeaponAction, Suppression, ProjectileImpact, Damage, Flashbang, SquadCallout, MatchPhase), interfaces declared in Core, or push-style setters.

Player rig — one writer per transform

```
Player            CharacterController, PlayerRig, Locomotion (yaw), StanceAndLean,
                  HeadbobAndBreathing, MantleAndVault (probes and path only)
└ BodyPivot       StanceAndLean: torso height, prone ground pitch
  └ LeanPivot     StanceAndLean: lean roll (±18°)
    └ HeadPivot   StanceAndLean: eye offset (stance + mount height)
      └ PitchPivot    Locomotion: look pitch
        └ HeadMotion  Headbob/Breathing: bob, landing, slide/dive/climb posture, sway
          ├ CameraOffset      PhysicalRecoilEngine: damped camera concussion
          │ └ PlayerCamera    ADS controller writes field of view (and shot kick)
          └ ViewmodelRoot     layer FirstPersonViewmodel
            ├ InertiaPivot    springs, free aim, trail, postures, sway, compressed ready, tuck
            │ └ RecoilPivot   per-shot kick about the stock
            │   └ ActionPivot reload, magazine check, bolt, holster/draw, inspect, climb carry
            │     └ AdsPivot  hip ⇔ solved sight alignment → weapon model
            └ Arms            two-bone IK to the weapon's grip sockets
```

Layers

Layer	#	Use
Player	8	Player and AI capsules
FirstPersonViewmodel	9	Weapon/arms; collides with nothing
WorldGeometry	10	Level geometry and door leaves; ballistics, tuck, lean
Hitbox	11	Per-bone damage colliders
Interactable	12	Interaction targets
Debris	13	Casings, magazines, grenades

Engineering rules

- Zero allocation on hot paths: pools for casings, magazines, decals, emitters, grenades; non-allocating physics; batched bullet raycasts.
- Domain-reload safe: statics reset at SubsystemRegistration.
- SI units everywhere; imperial only as authoring fields.
- Data-driven ScriptableObjects (`*SO`); models bind through sockets, never hardcoded paths.
- Bullets always spawn from the `MuzzleSocket` along the bore.
- No stubs or TODOs anywhere.

The repository's `CLAUDE.md` is the authoritative engineering guide and lists every contract between systems.

17 Content pipeline and tests

Models

- **Meshy specs**: 33 models (weapons, parts, equipment, props, operator gear) each have an id, category, prompt, texture prompt and fit in `MeshyAssetManifest`. Prompts are in `Art/Models/MeshyPrompts.md` and `Tools/FPS Builder/Meshy/Asset Queue`.
- **Drop folders**: a model exported from meshy.ai into `Art/Models/<Category>/<Id>/` (FBX/OBJ + PBR textures) wins on the next builder run. Import builds `<Id>_HDRP.mat`, packs metallic/roughness/AO into a mask map and fits the model to its part frame (long axis forward, uniform scale; `fit.json` overrides the orientation). Sockets on imported parts come from measured end slices.
- **Meshy API**: only used when `MESHY_API_KEY` is set, and bills separate API credits; the intended workflow is meshy.ai web exports with the subscription.
- **Procedural fallback**: high-detail geometry from `MeshKit` (chamfered boxes, lathes, extrusions, rails, tori), generated surface finishes, static parts merged per finish and moving parts separate; one mesh asset per piece, overwritten in place so references survive rebuilds.
- **Importing a real weapon**: select an FBX/GLTF → builder 6. It wraps the model under a `WeaponModelRig`, converts materials to HDRP/Lit, strips colliders, guesses sockets and magazine renderers and creates `<Name>_Data`. The rig inspector lists validation issues with fix buttons; place sockets with colour-coded gizmos and add a `WeaponSight` to optics.
- **Everything else**: soldiers, grenades, voices, scenes, NavMeshes and map schematics come from builders 5, 7 and 8; audio from `AudioSynth` with the override folders.

Tests (PlayMode, NUnit + InputTestFixture)

Suite	Covers
Core, Locomotion, Movement	Event bus, pooling, springs, IK, paces, stances, lean, slide, stamina; vault/mantle classification, jump, slide cancel, compressed ready, inspect, camera kick, hand tremor, doors
Weapon, WeaponController, NewWeapons	Recoil, ADS, PiP, chamber tracking, fire modes, staged and interrupted reloads, magazine checks, swaps; battle rifle, sidearm slide, weapon racks
Ballistics, Effects, Acoustics	Drag, penetration, ricochet, armour, near misses, decals, muzzle VFX; synthesis, room tails, overrides, Mach-cone cracks, low-ammo ping, gear foley
Photoreal, ModelPipeline	World look, clouds, interiors, weathered materials, edge wear, scope lens, sparks and dust, muzzle shock; mesh winding, mask packing, model fitting, Meshy manifest, detailed models
Gunsmith	Stat maths, measured inertia, socket moves, live refits, devices
AIPerception, AITactics	Sight, light, hearing, smoke; cover, suppression, flanking, room clearing, frag flushing
Equipment, Gameplay	Cooking, frags, stun, smoke; all three modes, spawns, objectives; the Raid opening unseen from every start
Armory, FrontEnd, CombatFeedback	Profiles, workbench, settings, looting; difficulty, rebinding, map schematics, deploy and return; hit classification, attribution, hit-marker modes
Footstep, ViewmodelPolish	Step timing and loudness by pace; rack weapons, view poses on screen, bolt stroke off the sights, scope exposure
RenderCapture (opt-in)	PNG captures of the range, modes and armory

Latest full run: **165 total, 160 passed, 0 failed, 5 skipped** (the render captures, which only run with `FPS_CAPTURE_DIR` set). New non-trivial logic ships with a test in `Assets/_Project/Tests`.

> **Art status.** Every model is high-detail procedural geometry with weathered PBR finishes and procedural motion (no animation clips). Meshy exports dropped into the drop folders replace them without code changes.

18 Known issues and workarounds

Issue	Detail and handling
Meshy models not generated yet	The manifest lists 33 models, but `Art/Models` holds no exports, so every model is still procedural. Generate them on meshy.ai with the subscription, drop the FBX/OBJ and textures into `Art/Models/<Category>/<Id>/`, then rerun builders 3–8.
Builder 6 needs a selected model	`6. Build Weapon From Selected Model` works on the FBX/GLTF selected in the Project window; with nothing selected it only logs an error. It is not part of the run-in-order chain.
Gamepad Interact is A hold	A tap now jumps and climbs, so interacting on the gamepad needs a 0.3 s hold of A. Keyboard F is unchanged.
No CI yet	Tests and player builds run locally in batch mode. CI (for example GameCI on GitHub Actions) needs a Unity license stored as a repository secret.
HDRP 17.6 fog bug (editor only)	Entering Play mode resets HDRP's local volumetric fog manager after the pipeline allocated its buffers, so every frame threw "Render Graph Execution error" in the fog voxelization pass. Fixed by `VolumetricFogBufferRepair` (commit `1b05b0a`), which re-allocates the buffers before the first scene loads. Built players are unaffected. Remove it once HDRP ships a fix.
Editor input needs Game view focus	Keyboard and mouse only reach the game after clicking inside the Game view (Input System default). Simulated-input tests confirm movement and firing work in every scene.
Harmless editor log noise	An `ArgumentOutOfRangeException` from Unity Search indexing appears once at editor start-up; it is Unity-internal.
Synthesized audio and TTS voices	Gunfire and foley are synthesized; callouts are generated with Windows speech synthesis (silent when built elsewhere until clips are assigned). Replace with recordings through the override folders and voice profiles — no code changes.
Workbench scope	Only weapons with attachment mounts offer parts; the marksman rifle and sidearm are fixed builds.
Upscalers	DLSS and FSR2 depend on hardware/platform support and fall back to TAAU.
Single-player	No networking. Rebinding covers keyboard bindings only.
Housekeeping	Build output lives outside the repo (`Desktop\Claude of duty Build`). A second, stale clone exists at `Desktop\Mage Match\Claude-of-duty` (Milestone 1 only) — don't work there. The editor leaves `HDRPDefaultResources` assets and `ProjectSettings/PackageManagerSettings.asset` modified or untracked; review them before committing.

19 Recommended next steps

1. **Art pass**: generate the 33 Meshy models on meshy.ai from the ready prompts, drop them into `Art/Models` and rerun the builders; then character models driven by the existing procedural IK, and environment kits replacing the blockouts.
2. **Audio pass**: recorded gunshots (close, distant, suppressed), mechanisms, impacts, explosions and voice lines through the override folders and voice profiles.
3. **CI**: GitHub Actions (GameCI) running the PlayMode suite and builder 9 on every push, with the Unity license as a secret.
4. **Content**: more weapons (SMG, shotgun), parts for the battle rifle and sidearm (suppressors, pistol optic), and maps built for each mode.
5. **Gamepad**: gamepad rebinding and a dedicated Interact button so A no longer needs a hold.
6. **Performance**: profile the photoreal look (SSGI, volumetrics, PiP) on mid-range GPUs and add quality presets to the Video tab.
7. **Release hygiene**: remove the HDRP fog workaround when Unity fixes the bug; version the player builds.